

Updated lime requirement recommendations for Oregon

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New research from the OSU Soil Fertility Program provides improved lime rate recommendations for both Sikora and SMP buffer methods, including higher pH targets up to 7.2.

Twenty-four acidic soils from across the state were tested using laboratory lime incubation trials to determine the lime needed to reach target soil pH values ranging from 5.6 to 7.2. This work led to updated lime requirement estimates for each target.

Recommendations are provided for both Sikora and SMP buffer methods (Tables 1 and 2). Lime recommendations for additional pH targets between those listed, along with the supporting equations for all pH targets, are available upon request in Excel format (amber.moore@oregonstate.edu).

Table 1. Updated OSU lime recommendations for Oregon based on the Sikora buffer pH

Sikora lime requirement test value	Amount of lime to apply to attain target soil pH (t/a)				
	Target pH of 5.6	Target pH 6.0	Target pH 6.4	Target pH 6.8	Target pH 7.2
6.9	0.0	0.0	0.5	0.8	1.0
6.8	0.0	0.4	0.8	1.2	1.6
6.7	0.2	0.7	1.1	1.6	2.2
6.6	0.4	0.9	1.5	2.1	2.7
6.5	0.7	1.2	1.8	2.5	3.3
6.4	0.9	1.5	2.1	2.9	3.8
6.3	1.1	1.7	2.5	3.3	4.4
6.2	1.3	2.0	2.8	3.7	4.9
6.1	1.5	2.3	3.1	4.2	5.5
6.0	1.7	2.5	3.5	4.6	6.0

	Amount of lime to apply to attain target soil pH (t/a)				
Sikora lime requirement test value	Target pH of 5.6	Target pH 6.0	Target pH 6.4	Target pH 6.8	Target pH 7.2
5.9	1.9	2.8	3.8	5.0	6.6
5.8	2.2	3.1	4.1	5.4	7.1
5.7	2.7	4.0	5.5	7.2	9.9
5.6	3.1	4.5	6.2	8.3	11.4
5.5	3.5	5.1	7.0	9.3	12.9
5.4	3.8	5.6	7.7	10.3	14.3
5.3	4.2	6.2	8.5	11.3	15.8
5.2	4.6	6.7	9.2	12.3	17.3
5.1	5.0	7.3	10.0	13	18.7
5.0	5.3	7.8	10.7	14	20.2
4.9	5.7	8.4	11.5	15	21.7
4.8	6.1	8.9	12.2	16.3	23.2

Recommended lime application rates are based on the use of 100-score lime and a 6-inch soil sampling depth. For example, if your Sikora buffer value is 6.0 and your target soil pH is 5.6, apply lime at the 1.7 t/a rate.

If the target soil pH has already been achieved, a lime application may not be necessary.

Lime rates reflect the minimum needed; higher rates may be necessary based on field-specific goals.

Table 2. Updated OSU lime recommendations for Oregon based on the SMP buffer pH

SMP lime requirement test value	Lime to apply to attain target soil pH (t/a)				
	Target pH 5.6	Target pH 6.0	Target pH 6.4	Target pH 6.8	Target pH 7.2
6.9	0.0	0.4	0.8	1.2	1.5
6.8	0.2	0.6	1.1	1.6	2.0
6.7	0.4	0.9	1.4	1.9	2.5
6.6	0.6	1.1	1.7	2.3	3.1
6.5	0.8	1.4	2.0	2.7	3.6
6.4	1.0	1.6	2.3	3.1	4.1
6.3	1.2	1.9	2.6	3.5	4.6
6.2	1.4	2.1	3.0	3.9	5.2

SMP lime requirement test value	Lime to apply to attain target soil pH (t/a)				
	Target pH 5.6	Target pH 6.0	Target pH 6.4	Target pH 6.8	Target pH 7.2
6.1	1.6	2.4	3.3	4.3	5.7
6.0	1.8	2.6	3.6	4.7	6.2
5.9	2.1	3.2	4.3	5.8	7.8
5.8	2.5	3.7	5.1	6.8	9.3
5.7	2.9	4.3	5.9	7.9	10.9
5.6	3.3	4.9	6.7	8.9	12.4
5.5	3.7	5.5	7.5	10.0	13.9
5.4	4.1	6.0	8.3	11.0	15.4
5.3	4.5	6.6	9.0	12.1	16.9
5.2	4.9	7.2	9.8	13.1	18.4
5.1	5.3	7.8	10.6	14.1	19.9
5.0	5.7	8.3	11.4	15.2	21.4
4.9	6.1	8.9	12.2	16.2	22.9
4.8	6.5	9.5	12.9	17.3	24.5

Recommended lime application rates are based on the use of 100-score lime and a 6-inch soil sampling depth. For example, if your SMP buffer value is 6.0 and your target soil pH is 5.6, apply lime at the 1.8 t/a rate.

If the target soil pH has already been achieved, a lime application may not be necessary.

Lime rates reflect the minimum needed; higher rates may be necessary based on field-specific goals.

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Soil Fertility

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